Amar Prakash Pandey

Senior Data Engineer

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Summary

Senior Data Engineer with **8+ years** of experience **designing scalable, high-performance data architectures**, **ELT/ETL pipelines**, and **large-scale batch processing systems**. Proficient in handling **petabyte-scale datasets** with **Apache Spark, Databricks, Airflow, and dbt**, backed by strong **SQL, Python, and Scala**. Skilled in building **cloud-native data solutions** on **AWS and GCP**, with a focus on **performance optimisation, cost efficiency, and data quality**.

Contributor to **Apache Airflow**; past maintainer at **CRI-O and DuckDuckGo**. Writes about **Spark internals and performance tuning** at amarpandey.me/blog

Experience

Citi — Pune, India

Senior Data Engineer — June 2025 – Present

- Cut Spark job resource usage by **up to 90%** by root-causing configuration issues in failing, over-provisioned production jobs; built Spark log and Autosys JIL analyzers that speed up diagnosis team-wide.
- Built **var-compare** from scratch, a daily validation tool that reconciles Market Risk numbers before reports are published.
- Led the **Maven → Gradle** migration of the trading-reports mono-repo: live in 2 weeks with **zero incidents**, **~40% faster builds**, and vulnerable dependencies removed.
- Initiated test automation on a zero-test ~20K-LOC codebase, reaching **60% coverage**; it became a team-wide mandate (80% minimum in CI). Codified team conventions as **Claude Code skills** so AI-generated code stays consistent across the team.

Sahaj Software — Pune, India

Senior Data Engineer — March 2021 – June 2025

Developed and maintained end-to-end batch pipelines to process audience and frame-level data for out-of-home (OOH) advertising, enabling accurate and timely delivery of insights to campaign optimisation and analytics teams. Owned data quality checks and pipeline performance end-to-end for effective ad targeting and impact measurement.

- **Designed and optimised ETL pipelines** with **Apache Spark (PySpark) on Databricks**, processing terabytes of data and executing large-scale joins to enrich advertisement analytics.
- **Enhanced Spark job performance by 50%** through advanced shuffle partition tuning and resource optimisation, significantly reducing processing time.
- **Implemented and managed** workflow orchestration with **Apache Airflow**, ensuring efficient sequencing, automated retries, and SLA compliance for batch processing.
- **Cut S3 storage costs by 60%** by implementing efficient data lifecycle policies and optimising file formats for cost-effective cloud storage.
- Automated deployment of **Airflow DAGs**, **Apache Spark/PySpark jobs**, and **microservices** using **GitLab CI/CD**, built with **Cursor AI** to deliver the setup in a fraction of the estimated time with broader test and edge-case coverage, eliminating manual errors and ensuring seamless promotion across dev and production environments.

Tata Consultancy Services — Chennai, India

System Engineer — December 2017 – February 2021

Developed a **pricing platform** that streamlined and automated pricing updates for thousands of components. My contributions focused on **optimizing workflows, improving system reliability, and enhancing deployment efficiency**.

- **Developed APIs** for Cummins' pricing platform using **Java, Spring Boot, and MySQL**, enabling real-time pricing updates for thousands of components.
- **Automated pricing workflows**, reducing update time from **one day to near real-time**.
- **Migrated Oracle trigger and function logic** to a service layer for **better logging and maintainability**.
- Automated deployment using CI/CD pipelines with Git, **Jenkins, and GitLab**, improving efficiency and reliability.

Skills

Programming Languages

Java, Python, Scala, SQL

Big Data Technologies

Apache Spark, PySpark, Databricks, Apache Hadoop

ETL Tools & Orchestration

Apache Airflow, dbt (Data Build Tool)

Cloud Technologies

AWS S3, ECS, ECR, Lambda, RDS, AWS EMR

Databases

PostgreSQL, Oracle

DevOps

Git, Docker, Terraform, Jenkins, GitLab CI/CD, Github

Projects

spot-optimizer January 2025

Built and published an **open-source Python library on PyPI** that replaces in-house logic for **AWS Spot Instance selection** and recommends the most stable, cost-effective instances based on cores, memory, region, and **EMR version compatibility**.

geohashviz.com July 2024

GeohashViz is a web-based visualization tool that helps users explore **geohashes** interactively. It allows users to encode locations into geohashes, visualize different precision levels, and understand spatial hierarchies

Finger Detection and Tracking April 2017

A computer vision project (**400+ GitHub stars**) that detects and tracks fingers in real time using **OpenCV and Python**, enabling gesture-based interactions from a live camera feed.

Awards

Senior Data Engineer March 2024

Recognized for exceptional contributions at **Sahaj.ai**, receiving multiple commendations for assisting projects, building useful tools, and consistently going above and beyond.

System Engineer June 2019

Cummins

Awarded Star Performer of the Year at **TCS** for outstanding performance on the **Cummins** project.

Software Developer August 2017

Google Summer of Code

Completed Google Summer of Code (2017), contributing to the **Teammates project** at the **National University of Singapore**.